ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

## PAPER 1 · PRACTICE A

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: computer literacy (hardware, software, files, spreadsheets, Internet and safety), problem solving and algorithms, flowcharts and pseudocode, programming basics (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Algorithms may be written in pseudocode, drawn as flowcharts or written in a programming language you have studied, unless a specific method is asked for.
5. Write neatly and legibly.

## SECTION A

## QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

**1.1** Which ONE of the following is an output device?

A Keyboard
B Mouse
C Webcam
D Printer

**1.2** In a spreadsheet, the cell in column C and row 5 is …

A C5.
B 5C.
C R5C.
D CC5.

**1.3** The spreadsheet formula that adds the values in cells B2 to B10 is …

A =ADD(B2-B10)
B SUM B2 to B10
C =SUM(B2:B10)
D =TOTAL(B2;B10)

**1.4** The keyboard shortcut to copy selected text in Windows is …

A Ctrl + V.
B Ctrl + X.
C Ctrl + C.
D Ctrl + Z.

**1.5** Which symbol is used for input or output in a flowchart?

A Parallelogram
B Diamond
C Oval
D Circle

**1.6** A named memory location used to store a value in a program is a …

A loop.
B function.
C comment.
D variable.

**1.7** Which data type stores a whole number such as 25?

A String
B Integer
C Boolean
D Float

**1.8** In the Arduino language, the function that runs over and over again is …

A setup().
B main().
C begin().
D loop().

**1.9** Software that controls the hardware and runs other programs is the …

A word processor.
B operating system.
C web browser.
D spreadsheet.

**1.10** The result of 10 > 5 in a program is …

A False.
B 15.
C True.
D 5.

(10 × 1) [10]

## QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (2.1–2.10) in the ANSWER BOOK.

**2.1** RAM keeps its data when the computer is switched off.

**2.2** A file extension such as .docx shows the type of file.

**2.3** Spreadsheet formulas start with an equals sign (=).

**2.4** An algorithm must have a clear end.

**2.5** A diamond in a flowchart represents a process (calculation).

**2.6** Ctrl + Z undoes the last action.

**2.7** A WHILE loop repeats as long as its condition is true.

**2.8** Programs written in a high-level language must be translated before the computer can run them.

**2.9** It is safe to share your password with friends.

**2.10** A USB flash drive is a secondary storage device.

(10 × 1) [10]

## QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 3.1 | Physical parts of a computer | A | algorithm |
| 3.2 | Programs that run on a computer | B | hardware |
| 3.3 | Container used to organise files | C | microcontroller |
| 3.4 | Ready-made formula in a spreadsheet, e.g. AVERAGE | D | selection |
| 3.5 | Step-by-step plan to solve a problem | E | loop |
| 3.6 | Repeating a set of instructions | F | function |
| 3.7 | Choosing between options with IF | G | development environment |
| 3.8 | A mistake in a program | H | software |
| 3.9 | Small computer on a board used to control robots, e.g. Arduino | I | bug |
| 3.10 | Program used to write and upload code, e.g. Arduino IDE | J | folder |

(10 × 1) [10]

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1** Name TWO input devices, TWO output devices and TWO storage devices. (6)

**4.2** Explain the difference between saving a file on the computer's hard drive and saving it in the cloud. Give ONE advantage of each. (4)

**4.3** Study the spreadsheet showing the parts bought for a robot project.

| | A | B | C | D |
|---|---|---|---|---|
| **1** | **Part** | **Quantity** | **Unit price (R)** | **Total (R)** |
| **2** | DC motor | 2 | 85 | |
| **3** | Ultrasonic sensor | 1 | 60 | |
| **4** | Arduino Uno | 1 | 250 | |
| **5** | Wheels | 4 | 30 | |
| **6** | **Grand total** | | | |

4.3.1 Write the formula for cell D2. (2)

4.3.2 Write the formula for cell D6. (2)

4.3.3 Calculate the value that will appear in cell D6. (2)

4.3.4 Write a formula to find the most expensive unit price. (2)

**4.4** Give FOUR rules for staying safe online. (4)

**[22]**

## QUESTION 5

**5.1** Write an algorithm (numbered steps in plain language) for a robot that must cross a room, avoiding obstacles, and stop at the far wall. (5)

**5.2** Draw a flowchart for a program that reads a temperature from a sensor and switches on a fan if the temperature is above 30 °C, otherwise switches it off. The program must repeat continuously. (8)

**5.3** Identify the input, processing and output in QUESTION 3.2. (3)

**[16]**

## QUESTION 6

**6.1** Study the pseudocode below and answer the questions.

```
1  count ← 1
2  total ← 0
3  WHILE count <= 5
4     total ← total + count
5     count ← count + 1
6  ENDWHILE
7  OUTPUT total
```

6.1.1 Complete a trace table showing count and total. (5)

6.1.2 What is the output? (1)

6.1.3 What would the output be if line 3 were changed to WHILE count < 5? (2)

**6.2** Write pseudocode that asks for a robot's battery voltage and displays "Charge now" if the voltage is below 6,5 V, "Battery low" if it is from 6,5 V up to 7 V, and "OK" otherwise. (6)

**6.3** Study the Arduino code below.

```
void setup() {
  pinMode(13, OUTPUT);
}
void loop() {
  digitalWrite(13, HIGH);
  delay(500);
  digitalWrite(13, LOW);
  delay(500);
}
```

6.3.1 Describe what this program does to an LED connected to pin 13. (3)

6.3.2 How many times per second does the LED switch on? (1)

6.3.3 What change would make the LED flash twice as fast? (2)

**6.4** Give TWO reasons for adding comments to program code. (2)

**[22]**

## QUESTION 7

**7.1** A student saves the robot project report as *Report final final2.docx* in the Downloads folder. Give TWO problems with this, and suggest a better file name and folder location. (4)

**7.2** Name FOUR features of a word processor that make a project report look professional. (4)

**7.3** Explain why you should back up your project files, and give ONE backup method. (2)

**[10]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**